

These are notes for Malek Abdesselam's Functional Analysis class for Fall 2026.

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Motivation

Consider a space V of (sufficiently nice) functions $f: \mathbb{R} \rightarrow \mathbb{R}$. Then, similar to how a vector in $\mathbb{R}^n$ can be denoted as

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may view the function $f: \mathbb{R} \rightarrow \mathbb{R}$ as a “column vector” with uncountable index. That is, $f = (f(x))_{x \in \mathbb{R}}$. If $L: V \rightarrow V$ is a linear map, then an analogue of the finite-dimensional matrix representation can be viewed as a kernel that f is integrated against; that is,

$$(Lf)(x) = \int_{\mathbb{R}} K(x, y) f(y) dy.$$

In the special case that $L = \text{id}_V$, then the kernel $K(x, y)$ is given by the Dirac delta distribution, $K(x, y) = \delta(x - y)$. The space of these distributions is the primary object of study in Fourier analysis.

Locally Convex Topological Vector Spaces

Recall that if X is a set, then $\mathcal{T} \subseteq \mathcal{P}(\mathcal{P}(X))$ is a topology on X if it satisfies:

- $\emptyset, X \in \mathcal{T}$;
- $U_1, U_2 \in \mathcal{T}$ implies $U_1 \cap U_2 \in \mathcal{T}$;
- $\{U_i : i \in I\} \subseteq \mathcal{T}$ implies $\bigcup_{i \in I} U_i \in \mathcal{T}$.

Definition: Definitions

We say $(V, \mathcal{T})$ is a *topological vector space* (over $\mathbb{K}$) if V is a $\mathbb{K}$ -vector space whose topology is such that the maps

$$\begin{aligned} V \times V &\rightarrow V \\ (u, v) &\mapsto u + v \end{aligned}$$

and

$$\begin{aligned} \mathbb{K} \times V &\rightarrow V \\ (\lambda, v) &\mapsto \lambda v \end{aligned}$$

are continuous with respect to the product topology. Note that $\mathbb{K}$ is equipped with its traditional topology.

The morphisms of topological vector spaces are the *continuous* linear maps. We say $V \cong W$ are isomorphic as topological vector spaces if there is $\varphi: V \rightarrow W$ that is bijective, linear, and continuous, and φ^{-1} is also continuous.

Definition: Let V be a $\mathbb{K}$ -vector space. We say $\rho: V \rightarrow [0, \infty)$ is a *seminorm* if

$$\begin{aligned}\rho(v + w) &\leq \rho(v) + \rho(w) \\ \rho(\lambda v) &= |\lambda| \rho(v)\end{aligned}$$

for any $v, w \in V$ and $\lambda \in \mathbb{K}$.

Remark: The only difference between a seminorm and a norm is that $\rho(v) = 0$ need not imply that $v = 0$. A norm is a seminorm.

We will let $\text{SN}(V)$ be the set of all seminorms of V .

Definition: Let V be a $\mathbb{K}$ -vector space. Let $\mathcal{N} \subseteq \text{SN}(V)$. Define $\mathcal{T}(\mathcal{N})$ to be a topology with basic open sets $U(x, F, \varepsilon) \subseteq V$ given by

$$U(x, F, \varepsilon) = \{y \in V : \rho(y - x) < \varepsilon \text{ for all } \rho \in F\}$$

for any $x \in V$, $F \subseteq \mathcal{N}$ finite, and $\varepsilon > 0$.

We observe that the subsets of the form $U(x, \{\rho\}, \varepsilon)$ as ρ ranges across $\mathcal{N}$ forms a subbase for the topology of V .

Proposition: If $F \subseteq \mathcal{N}$ is finite, $\varepsilon > 0$, and $x \in V$, then

$$U(x, F, \varepsilon) \in \mathcal{T}(\mathcal{N}).$$

Proof. Suppose $y \in U(x, F, \varepsilon)$. Our goal is to find $\eta > 0$ such that $U(y, F, \eta) \subseteq U(x, F, \varepsilon)$. Suppose $z \in U(y, F, \eta)$. If $\rho \in F$, then

$$\begin{aligned}\rho(z - x) &\leq \rho(z - y) + \rho(y - x) \\ &< \eta + \rho(y - x).\end{aligned}$$

Therefore, we will define η to be small enough such that $\eta + \max_{\rho \in F} \rho(y - x) < \varepsilon$. This gives the desired result. $\square$

Definition: A topological space is called a locally convex topological vector space (LCTVS) if the topology is generated by a family of seminorms $\mathcal{N}$ with the property that

$$\bigcap_{\rho \in \mathcal{N}} \{v : \rho(v) = 0\} = \{0\}.$$

A family $\mathcal{N}$ of seminorms that satisfies this property is said to be *separating*.

Example: If H is a Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$, then the seminorms $v \mapsto |\langle v, e_i \rangle|$ are separating.

Theorem: If V is a $\mathbb{K}$ -LCTVS and $\mathcal{N}$ is a collection of seminorms that defines the topology on V , then $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff if and only if $\mathcal{N}$ separates the points of V .

Proof. Assume $\mathcal{N}$ separates the points of V . Suppose $v \neq w$ are distinct elements of V . Then, since $v - w \neq 0$, there is some $\rho \in \mathcal{N}$ such that $\rho(v - w) = \varepsilon_0 > 0$. Consider the finite set $F = \{\rho\}$, and choose some $0 < \varepsilon < \varepsilon_0/2$. Then, we claim that the basic open sets $U(v, F, \varepsilon)$ and $U(w, F, \varepsilon)$ are disjoint. This follows from the fact that if $x \in U(v, F, \varepsilon)$, then $\varepsilon_0 = \rho(v - w) \leq \rho(v - x) + \rho(x - w)$, so that $\rho(w - x) \geq \varepsilon_0 - \varepsilon > \varepsilon_0 - \varepsilon_0/2 > \varepsilon$, yielding the desired result.

Now, suppose $\mathcal{N}$ does not separate the points of V . Then, there are some distinct $v \neq w$ such that for all $\rho \in \mathcal{N}$, we have $\rho(v - w) = 0$. In particular, observe that for all finite subsets $F \subseteq \mathcal{N}$ and all $\varepsilon > 0$, we have that w is contained in $U(v, F, \varepsilon)$, meaning that there are no basic open sets that separate v and w . Since any open subset of V contains a basic open subset, it follows that for any open subsets $v \in U$ and $w \in U'$, we

have some finite F such that $U(v, F, \varepsilon) \subseteq U$, but this also has $w \in U(v, F, \varepsilon)$, implying that $U \cap U' \neq \emptyset$. Thus, V is not Hausdorff. $\square$

Examples

Remark: If we remove the assumption that the family $\mathcal{N}$ is separating, then the topology induced by the trivial seminorm $\rho: V \rightarrow [0, \infty)$ is the indiscrete topology on V . Similarly, there is a family of seminorms that induces the discrete topology on V .

- (i) Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Examples of these include Banach spaces such as $L^p(X, \Sigma, \mu)$, where Σ is a σ -algebra and μ is a positive measure on the measurable space (X, Σ) , and $p \in [1, \infty]$.

There is a metric on this normed space $d(x, y) = \|x - y\|$, which gives rise to the metric topology $\mathcal{T}(d)$. We have $U \in \mathcal{T}(d)$ if and only if $U \subseteq V$ is such that for any $x \in U$, there exists $\varepsilon > 0$ such that

$$\{y \in V : d(x, y) < \varepsilon\} =: U(x, \varepsilon) \subseteq U.$$

This is a locally convex space with seminorm $\mathcal{N} = \{\|\cdot\|\}$. There are two possibilities for finite subsets $F \subseteq \{\|\cdot\|\}$, meaning that if $F = \emptyset$, then $U(x, F, \varepsilon) = U(x, \emptyset, \varepsilon) = V$, while if $F = \{\|\cdot\|\}$, then

$$\begin{aligned} U(x, F, \varepsilon) &= \{y \in V : \rho(x - y) < \varepsilon \text{ for all } \rho \in F\} \\ &= \{y \in V : \|x - y\| < \varepsilon\} \\ &= U(x, \varepsilon). \end{aligned}$$

- (ii) Define $s(\mathbb{N}, \mathbb{K})$ to be the space of sequences $x = (x_n)_{n \geq 0}$ in $\mathbb{K}^{\mathbb{N}}$ such that for all $k \geq 0$, $\sup_{n \geq 0} \langle n \rangle^k |x_n| < \infty$, where $\langle n \rangle = \sqrt{1 + n^2}$ is the bracket. This is the space of *rapidly decaying sequences* in $\mathbb{K}$. In essence, elements of $s(\mathbb{N}, \mathbb{K})$ decay faster than any inverse power of the index n .

We define the seminorms

$$\rho_k(x) = \sup_{n \geq 0} \langle n \rangle^k |x_n|.$$

In fact, the $\{\rho_k : k \in \mathbb{N}\}$ are norms. Let

$$\mathcal{N} := \{\rho_k : k \in \mathbb{N}\}$$

We obtain a locally convex on $s(\mathbb{N}, \mathbb{K})$ with topology induced by the family $\mathcal{N}$.

Later, we will see that $(s(\mathbb{N}, \mathbb{K}), \mathcal{T}(\mathcal{N}))$ is *not* a normed space. That is, there does not exist a norm $\|\cdot\|$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\{\|\cdot\|\})$.

- (iii) Take $V = C([0, \infty), \mathbb{R})$ to be the set of all real-valued continuous functions on $[0, \infty)$ with pointwise addition and scalar multiplication. This is a real vector space.

At first, we may think $\|f\| = \sup_{t \in \mathbb{R}} |f(t)|$ would define a norm, but this is not satisfactory since the functions in V are not necessarily bounded.

Therefore, we want to restrict to compact subsets. Take

$$\rho_k(f) := \sup_{0 \leq t \leq k} |f(t)|.$$

Notice that the ρ_k are not norms (use Urysohn's Lemma or your favorite piecewise function to construct one). However, the family $\mathcal{N} = \{\rho_k : k \geq 0\}$ is a separating family.

Consider the set of functions $V_0 = \{f \in V : f(0) = 0\}$. Then, $V_0 \leq V$ is a closed subset.

Proof. Consider the linear functional $\text{ev}_0: V \rightarrow \mathbb{R}$ given by $f \mapsto f(0)$. We claim that this linear functional is continuous.

Let $U \subseteq \mathbb{R}$ be open. Let $z \in U$, and let $\varepsilon > 0$ be such that $U(z, \varepsilon) := \{y \in \mathbb{R} : |z - y| < \varepsilon\}$ is contained in U . Let $f \in \text{ev}_0^{-1}(U(z, \varepsilon))$, and set $\varepsilon_0 = |z - f(0)|$. To show that $\text{ev}_0^{-1}(U(z, \varepsilon))$ is open, it suffices to find some seminorm ρ_k and some $\delta > 0$ such that $U(f, \rho_k, \delta) \subseteq \text{ev}_0^{-1}(U(z, \varepsilon))$.

In fact, we may choose $k = 1$ and $\delta = \varepsilon - \varepsilon_0$. Suppose $g \in U(f, \rho_1, \delta)$. Then, we have that

$$\begin{aligned} |g(0) - z| &\leq |g(0) - f(0)| + |f(0) - z| \\ &\leq \sup_{t \in [0,1]} |g(t) - f(t)| + \varepsilon_0 \\ &< \delta + \varepsilon_0 \\ &= \varepsilon. \end{aligned}$$

In particular, Thus, $U(f, \rho_1, \delta) \subseteq \text{ev}_0^{-1}(U(z, \varepsilon))$, so that $\text{ev}_0^{-1}(U(z, \varepsilon))$ is open as any element in this set admits a basic open neighborhood contained in the set. Therefore, ev_0 is continuous.

Since $V_0 = \text{ev}_0^{-1}(\{0\})$, it follows that V_0 is closed since $\{0\} \subseteq \mathbb{R}$ is closed. $\square$

We claim that the subspace topology on V_0 is given by $\mathcal{T}_0 = \mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$.

Proof. First, we will show that, given a subbasic open neighborhood $U = V_0 \cap U(x, \rho_k, \varepsilon)$ in the subspace topology of V_0 and any element y in this subbasic open neighborhood, there is an open neighborhood $V \in \mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$ such that $y \in V \subseteq U$.

If $y \in U$, we observe that $y \in U(x, \rho_k, \varepsilon)$. In particular, since these form a subbase for $\mathcal{T}(\{\rho_k : k \geq 0\})$, it follows that there is some finite collection $F = \{\rho_{i_j} : j = 1, \dots, m\}$ of seminorms and some $\delta > 0$ such that

$$U(y, F, \delta) = \bigcap_{j=1}^m U(y, \rho_{i_j}, \delta).$$

Since $U(y, \rho_{i_j}|_{V_0}, \delta) \subseteq U(y, \rho_{i_j}, \delta)$ for each j , it follows that we have

$$\bigcap_{j=1}^m U(y, \rho_{i_j}|_{V_0}, \delta) \subseteq V_0 \cap \left(\bigcap_{j=1}^m U(y, \rho_{i_j}, \delta) \right),$$

whence setting V to equal the left hand side yields the desired result. Therefore, $\mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$ refines the subspace topology $\mathcal{T}_0$.

In the reverse direction, observe that if $y \in U(x, \rho_j|_{V_0}, \varepsilon)$ in V_0 , then $y \in U(x, \rho_j, \varepsilon)$, so there is some $\delta > 0$ and finite $F = \{\rho_{i_j} : j = 1, \dots, m\}$ such that

$$\begin{aligned} U(y, F, \varepsilon) &= \bigcap_{j=1}^m U(y, \rho_{i_j}, \delta) \\ &\subseteq U(x, \rho_j, \varepsilon), \end{aligned}$$

but we notice then that $U(x, \rho_j, \varepsilon) \cap V_0 \subseteq U(x, \rho_j|_{V_0}, \varepsilon)$, so that $V_0 \cap U(y, F, \varepsilon) \subseteq U(x, \rho_j|_{V_0}, \varepsilon)$, so that $\mathcal{T}_0$ refines $\mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$. Thus, these topologies are equal. $\square$

We may consider $\mathcal{B}(\mathcal{T}_0)$ to be the Borel σ -algebra on V_0 . Brownian motions are a special class of probability measure on $(V_0, \mathcal{B}(\mathcal{T}_0))$. Specifically, they are the probability measures such that for every

$n \geq 0$, partition $0 < t_1 < t_2 < \dots < t_n$, the random variables

$$\begin{aligned} X_1 &= f \mapsto f(t_1) \\ X_2 &= f \mapsto (f(t_2) - f(t_1)) \\ X_3 &= f \mapsto (f(t_3) - f(t_2)) \\ &\vdots \\ X_n &= f \mapsto (f(t_n) - f(t_{n-1})) \end{aligned}$$

are independent centered Gaussian random variables with variances $t_1, t_2 - t_1, \dots, t_n - t_{n-1}$ for an arbitrary f .

Similar to the case of $s(\mathbb{N}, \mathbb{K})$, the space $(V_0, \mathcal{T}_0)$ is not a normed space.

- (iv) Consider the vector space $V = \mathbb{R}^{\mathbb{N}}$, which is the space of all real-valued sequences. Let $\mathcal{T}$ be the product topology on $\mathbb{R}^{\mathbb{N}}$. We are curious as to whether $(V, \mathcal{T})$ is a locally convex topological vector space.

Consider the function $d: V \times V \rightarrow [0, \infty)$ defined by

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min(1, |x_n - y_n|).$$

Then, we have that d is a distance metric, and $\mathcal{T} = \mathcal{T}(d)$.

Proof. We start by showing that if $x \in V$ is any element and $\varepsilon > 0$, then $U(x, \varepsilon)$ is open in $\mathcal{T}$. Translating, we take $x = 0$ (since the metric d is translation-invariant), so that we are now seeking to show that $U(0, \varepsilon)$ is open. This is equivalent to showing that if $x \in U(0, \varepsilon)$, then there is a finite collection of subbasic open sets $\pi_j^{-1}(U_j)$ whose intersection W contains x and is contained in $U(0, \varepsilon)$.

To start, we observe that there is some index n_0 such that

$$\sum_{n=n_0+1}^{\infty} 2^{-n} < \varepsilon/2.$$

We then select $U_j = (x_j - \varepsilon/4, x_j + \varepsilon/4)$ for each $j = 0, \dots, n_0$. It then follows that $x \in W$, and we have for any $y \in W$ that

$$\begin{aligned} d(x, y) &= \sum_{j=0}^{\infty} 2^{-j} \min(1, |x_j - y_j|) \\ &\leq \sum_{j=0}^{n_0} 2^{-j} \min(1, |x_j - y_j|) + \sum_{j=n_0+1}^{\infty} 2^{-j} \\ &< \sum_{j=0}^{n_0} 2^{-j} (\varepsilon/4) + \varepsilon/2 \\ &\leq \varepsilon, \end{aligned}$$

whence $x \in W \subseteq U(0, \varepsilon)$.

Now, consider a basic open set $W = \bigcap_{j=0}^{n_0} \pi_j^{-1}(U_j)$. Let $x \in W$. Let $\delta_0, \dots, \delta_{n_0}$ be such that $(x_j - \delta_j, x_j + \delta_j) \subseteq U_j$ for each $j = 0, \dots, n_0$, and set $\delta = 2^{-n_1} \min(\delta_0, \dots, \delta_{n_0})$, where n_1 is any integer strictly larger than n_0 .

Notice then that if $y \in U(x, \delta)$, then we have for each $j = 0, \dots, n_0$ that

$$|x_j - y_j| < 2^j \delta$$

$$\begin{aligned}
&\leq 2^{n_0} 2^{-n_1} \min(\delta_0, \dots, \delta_{n_0}) \\
&< \min(\delta_0, \dots, \delta_{n_0}) \\
&\leq \delta_j.
\end{aligned}$$

In particular, this means we must have $U(x, \delta) \subseteq W$.

Therefore, we have that $\mathcal{T}$ and $\mathcal{T}(d)$ generate the same topology on V . $\square$

In particular, we may consider the seminorms

$$\rho_n(x) = |x_n|,$$

and $\mathcal{N} = \{\rho_n : n \geq 0\}$. Then, we claim that $\mathcal{T} = \mathcal{T}(\mathcal{N})$.

| *Proof.* $\square$

In particular, V is not a normed space.

- (v) Consider $c_{00}(\mathbb{N}, \mathbb{K}) \cong \mathbb{K}[x]$ to consist of all the finitely supported sequences $(x_n)_{n \geq 0}$. This is a vector space. Consider the seminorms $\mathcal{N} = \text{SN}(V)$, and the locally convex space $(V, \mathcal{T}(\mathcal{N}))$. Then, this is a locally convex space that is not metrizable. The topology $\mathcal{T}(\mathcal{N})$ is known as the finest locally convex topology.

Notation

- The natural numbers are given by $\mathbb{N} = \{0, 1, 2, 3, \dots\}$.
- The strictly positive natural numbers are given by $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$.
- A multi-index is a tuple $\alpha \in \mathbb{N}^n$. We let

$$|\alpha| = \alpha_1 + \dots + \alpha_n.$$

If $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, then we will write

$$x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$$

and

$$\alpha! = \alpha_1! \dots \alpha_n!$$

for the corresponding values.

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is any function, then we will write

$$\partial^\alpha f = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}.$$

- If A is a set, we will write $\mathcal{P}(A)$ for the power set of A . If A and B are sets, then the set $B^A \subseteq \mathcal{P}(A \times B)$ is the set of all functions $f: A \rightarrow B$.

We will write $\mathcal{P}_{\text{fin}}(X)$ to consist of all finite subsets of X .

- If $x \in \mathbb{R}^n$ is given by $(x_1, \dots, x_n)$, we will define the bracket

$$\langle x \rangle := \sqrt{1 + x_1^2 + \dots + x_n^2}.$$

In the particular case of $n = 1$, we will write $\langle k \rangle = \sqrt{1 + k^2}$ for any $k \in \mathbb{N}$.

- If $n \in \mathbb{N}$, then

$$[n] = \{1, 2, \dots, n\},$$

where we have $[0] = \emptyset$.

- If A is a finite set, then $|A|$ is the cardinality of A .
- Generally we will denote by $\mathbb{K}$ for either $\mathbb{R}$ or $\mathbb{C}$.